22-S2-Q3

(a) method + procedure

(b) 400 $\begin{cases} 1 \\ -1 \end{cases}$

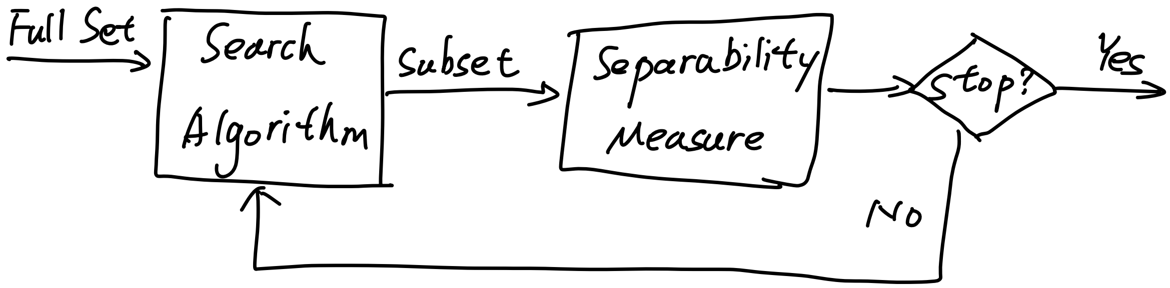
m_1 m_2 S_1 S_2

Fisher linear discriminane

(c) repeated k-fold

Solution ① The filter method

② procedure map



③ Search Algorithm can be

(1) Exhaustive search method

(2) Sequential forward selection

(3) Sequential backward elimination

④ Separability measures can be

(1) Mahalanobis distance-based separability measure

(2) Scatter-based separability measure

⑤ the stopping criterion could be

(1) number of features to be selected is reach

(2) separability measure no longer improves

(3) Others

⑥ Example

Sequential forward feature selection algorithm

(1) In step 1, consider each feather as a condidate

$$F^{(i)} = \{x_i\} \qquad i = 1, 2, \cdots n$$

evaluate the separability measure

$$J^{F(i)} = \frac{Tr(S_B^{F(i)})}{Tr(S_W^{F(i)})}$$

The feature that has the largest separability is selected as the first feature

$$k = \arg\max(J^{F(i)})$$

$$z_1 = x_k$$

(2) In step 2, consider each of the remaining $n-1$ features as the candidate for the second feature

$$F^{(i)} = \{z_1, x_i\}, \quad i \neq k$$

Evaluate the separability measure

$$J^{F(i)} = \frac{Tr(S_B^{F(i)})}{Tr(S_W^{F(i)})}$$

The feature that has the largest separability is selected as the second feature

(3) The selection process is continued until the stopping criterion is satisfied

(b) ① $S_w = S_1 + S_2$

$$w^* = S_w^{-1}(m_1 - m_2)$$

$$b^* = -\frac{(w^*)^T(m_1+m_2)}{2}$$

$$g(x) = w^T x + w_0$$

$$= (w^*)^T x + b^*$$

If $g(x) > 0$, classifying x to class 1

If $g(x) < 0$, classifying x to class -1

If $g(x) = 0$, x on the decision boundary

② $w^* = S_w^{-1}(m_1 - m_2)$

$$= \begin{bmatrix} -0.005344 \\ -0.01077 \\ -0.01089 \end{bmatrix}$$

$$b^* = -\frac{(w^*)^T(m_1+m_2)}{2}$$

$$= 0.001298$$

③ we denote $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$$g(x) = -0.005344 x_1 - 0.01077 x_2 - 0.01089 x_3 + 0.001298$$

(c)

In k-fold cross validation, the data set is divided into k- completely distinct or non-overlapping random partitions called folds. Each of the k folds is given an opportunity to be used as a held back test set, whilst all other folds collectively are used as a training dataset. A total of k models are fit and evaluated on the k hold-out test sets and the mean performance is reported.